

ENERGY INFRASTRUCTURE ECONOMICS

POWER • FUELS • SECURITY • TRANSITION

Economy Topic 31 | Current-status cutoff: 10 September 2026

1 ENERGY-SERVICE CHAIN

- Resource/import -> conversion -> network/storage
- -> distribution -> final energy -> useful service
- Evaluate delivered reliability and affordability.

2 ACCOUNTING FIREWALL

- MW/GW = power; MWh/GWh/BU = energy
- Capacity != generation != peak != supply != consumption
- Storage needs MW + MWh + duration + efficiency.

TRAP: always attach unit, date, denominator and system boundary.

3 SECURITY HEXAGON

- Availability + access + affordability
- + reliability + sustainability + resilience
- Independence means lower vulnerability, not autarky.

4 ELECTRICITY VALUE CHAIN

- Generation -> CTU/STU -> Grid-India/NLDC/RLDC/SLDC
- -> DISCOM/open access -> meter -> consumer
- Schedule -> deviation settlement -> ancillary response.

5 LAW + INSTITUTIONS

- Electricity Act, 2003: licensing, access, tariff, appeal
- CEA technical | CERC inter-state | SERC state/retail
- Draft 2025 amendment was not enacted at 10 Sep 2026.

6 TARIFF + DISCOM

- $\text{Tariff} = \text{power} + \text{network/loss} + \text{O\&M/finance} + \text{return}$
- Subsidy != cross-subsidy != regulatory asset
- FY25: AT&C 15.04%; ACS-ARR gap Rs 0.06/kWh.

TRAP: accounting improvement does not by itself prove cash discipline or service.

7 COAL ECONOMICS

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7 COAL ECONOMICS

- Mine -> grade -> rail -> stock -> generation -> ash/closure
- FY25 output 1,047.523 MT; FY26 imports 246.37 MT
- Linkage/SHAKTI != mine auction != e-auction.

TRAP: production, dispatch, receipt and generation are separate physical stages.

8 OIL + GAS SECURITY

- Upstream -> midstream -> downstream; PNGRB excludes production
- FY26 provisional dependence: crude 88.7%; gas 50.1%
- SPR is a finite buffer; diversify source, route and contract.

TRAP: reserve capacity does not establish fill level or days of cover.

9 RENEWABLE PORTFOLIO

- Solar daytime | wind seasonal | hydro flexible/hydrology-bound
- Auction -> award -> finance -> grid -> commission -> generation
- Low bid != retail tariff; no fuel cost != zero system cost.

10 GRID + STORAGE

- Forecast + transmission/GEC + flexibility + reserves

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- Forecast + transmission/GEC + flexibility + reserves
- + BESS/PSP + demand response -> reliable variable RE
- CEA need: 336 GWh by 2029-30; 411 GWh by 2031-32.

TRAP: storage is not generation; state power and energy ratings together.

11 POLICY INSTRUMENTS

- RPO/RCO = renewable obligation | ESO = storage obligation
- REC = renewable MWh | ESCert = energy saving
- Carbon credit = GHG performance; prevent double counting.

12 ANSWER + EVALUATION

- Define/date/unit -> trace chain + institution + economics
- -> test justice, environment, finance and import risk
- Verdict: reliable delivered energy at durable total system cost.